ForgeCare AI Project Explanation

Business problem, end to end workflow, model evidence and demonstration guide

Manufacturing and automotive theme | Group 4 AI and ML ILP | Prototype status as checked on 6 November 2026

ForgeCare AI is a working decision support prototype for small automotive component manufacturers. It links camera evidence, technician review, an explainable inspection queue, approved maintenance information and model experiments. Its central purpose is to help a limited team decide what to inspect next and record why. The current live demonstration inspects phones as accessible physical objects; the automotive research uses real assembly line images. Neither phone exercise establishes factory deployment readiness.

The project in one minute

A camera can reveal a possible surface issue, but a single image is often unclear and an alert alone does not say which task matters most. ForgeCare records fresh, quality checked photos; lets a technician correct a draft review; opens an inspection case; ranks cases using visible rules and available worker time; retrieves source linked guidance; and records what was actually found. Reviewed examples may later support local and federated model training when enough independent physical samples exist.

Current demonstration	What it proves	What it does not prove
Live phone photos and worker review	The capture, review, case and audit workflow can run on real images.	Reliable dent counts or accuracy on the iPhone and realme.
Renault pipe clip and cast part studies	Models were actually trained and evaluated on public manufacturing photos.	A safe, approved factory inspection model.
Two client Flower rounds	Separate data partitions trained compatible weights and combined them.	Cross factory privacy or independent site training.

1 Business problem and proposed solution

Small manufacturers often have more assets to check than available inspection minutes. Photos, maintenance history, work orders and manuals may live in separate places. A low quality frame can look like a defect; a good looking frame cannot establish internal safety. A supervisor also needs to know whether a camera is current, whether an issue has already been reviewed and whether the recommended procedure applies to that exact equipment model.

ForgeCare proposes one traceable chain from observation to outcome. The product is designed to reduce missed handoffs and make worker effort easier to direct. It does not calculate a failure probability, autonomously clear a part, shut down machinery or replace a qualified engineer. A later customer pilot must measure whether this workflow actually improves throughput, recheck burden and useful findings.

Pain point	ForgeCare response
Unclear or stale camera evidence	Display live status and quality checks; ask for recapture rather than creating a confident finding.
Too many competing alerts	Use versioned priority rules and available inspection minutes to build an explainable queue.
Unverified AI descriptions	Keep model boxes and generated wording editable; require technician confirmation before a finding becomes a saved label.

Pain point	ForgeCare response
Scattered maintenance guidance	Search approved, model scoped sources and show citations; withhold unsupported instructions.
Customers cannot pool all raw photos	Demonstrate weight aggregation with compatible local models; validate separate site operation before claiming it.

2 Who uses the system

- Technician: captures several views, checks image quality, corrects the review, confirms visible counts and records the actual inspection result.
- Supervisor: assigns work, reviews priority reasons, approves inspection instructions and monitors completion.
- Reviewer or administrator: manages assets, cameras, rules, knowledge sources and model approval; sees experiment and federation evidence.
- AI services: suggest possible visual regions and draft source linked information. They do not make the final operational decision.

3 Complete operating workflow

1. **Register the subject and camera.** Create the asset with its site, exact model, criticality and inspection interval. Map a camera to that asset and a stable physical sample ID. In this demonstration the subjects are an iPhone 14 Plus and a realme narzo 60 5G. The Samsung Galaxy S24 FE is a camera source, not a subject.
2. **Connect a truly live source.** The browser camera uses the device that opened the HTTPS page and needs browser permission. The Samsung IP Webcam can be reached from a laptop on the same Wi Fi through the local bridge and paired HTTPS relay. A server webcam source uses the host laptop instead. A stopped or missing source must not display old photos as live video.
3. **Capture distinct views.** Take front, back and edge photos under adequate light. Each capture records time, source, asset, session and physical sample. The application checks brightness and focus. A dark or blurred image remains evidence of a failed capture, but it cannot be called a clear normal result.
4. **Build an assisted review.** Select up to five usable photos from the same camera, session and physical sample. The local visual models produce tentative per photo suggestions and overlays. An optional Groq visual draft can describe what it appears to see in up to three photos. The technician compares views, edits the note and checks scratch or dent counts on the actual phone. The same mark seen twice is counted once.
5. **Save a human finding.** The technician chooses no visible issue, minor issue, significant issue or unclear. AI assisted values require explicit confirmation. Unclear evidence requests recapture. A clear saved review opens or reuses a work order for that capture, avoiding duplicate cases from repeated clicks.
6. **Prioritize and assign.** The queue uses asset criticality, repeated reviewed conditions and how overdue inspection is. It shows reasons, attention level, assignee and capacity fit. A supervisor can assign the case and approve instructions. Approval moves it to Ready to inspect; it has not yet been physically completed.
7. **Use cited guidance.** A worker can ask the Knowledge assistant about the selected phone or a manufacturing topic. It retrieves approved source passages and gives a conversational answer only when those passages support it. The source links remain visible for verification.
8. **Inspect and close.** A worker records the real finding, actual minutes and whether the physical subject was inspected. Saving that outcome changes the work order to Completed. Audit and reports preserve the action, time and responsible user.

9. **Learn from reviewed data.** Saved human labels can support later experiments. A six hour scheduled check attempts separate iPhone and realme training, but skips correctly when it lacks enough different acceptable and damaged physical phones. Research results are never silently promoted to live use.

4 How priority and capacity work

The starting rule weights are 40 percent asset criticality, 35 percent repeated reviewed condition and 25 percent inspection overdue. Each component is normalized to a value from zero to one; their weighted sum is shown on a zero to 100 urgency scale. A new rule version can change these weights, provided they still total 100 percent. The default planning estimate is 15 minutes per task until actual durations are available.

The score is an ordering aid, not a likelihood that a part will fail. The capacity view walks the ranked queue and selects tasks that fit the available minutes. For example, four 15 minute tasks fit a 60 minute slot; a longer task may be skipped if it does not fit. The team should inspect the reasons and override the order when operational context calls for it. The current demonstration has no measured factory duration model or verified economic return.

5 What each screen is for

Screen	Main purpose
Overview	Current activity and high level state of cameras, cases and services.
Live cameras	Fresh camera status, current frame and connection problems; old captures are not presented as a live feed.
Capture review	Several photos, model suggestions, editable technician note and confirmed visible condition.
Inspection queue	Work orders, priority reasons, assignment, approval, actual inspection and completion.
Assets and setup	Phone or equipment registry, camera mapping, physical sample, rule version and source setup.
Knowledge assistant	Cited answers from approved manuals and guidance, plus a bounded draft agent.
Models and experiments	Public dataset studies, held out examples, training candidates and curriculum comparisons.
Federation	Completed Flower round timelines, phone readiness, six hour attempts and client updates.
Reports, administration, audit	Business summaries, users and approvals, and a readable record of actions.

6 Model studies and what their numbers mean

The page contains separate research studies. A held out image is one reserved for testing that particular model. Defects caught are defective examples flagged for review. Defects missed are defective examples given no alert. Good items flagged create unnecessary rechecks. These counts are more useful to the plant team than an isolated ROC AUC number. The datasets and splits differ, so their scores are not a single league table.

Study and data	Held out business result	Interpretation
Renault AutoVI pipe clips, normal only PaDiM	67 of 71 staged defects caught; 67 of 98 good clips flagged.	Real automotive images, but too many false alerts for line control.
Renault AutoVI pipe clips, supervised Flower	19 of 21 staged defects caught; 30 of 60 good clips flagged.	Actual three round training on a different, labelled split. Also unsuitable for release.
Cast pump impellers, supervised Flower	96 of 117 defective parts caught; 23 of 78 good parts flagged; 21 defects missed.	Real manufactured pump parts, not vehicle specific. Recheck minutes are estimates.

Study and data	Held out business result	Interpretation
Kaggle phone screen cracks	66 of 68 cracked screens flagged; 2 of only 7 intact screens flagged.	Small intact test group; no iPhone or realme validation and no dent count.
Roboflow phone exterior boxes	Dents found 0 of 14; major scratches 9 of 24; small scratches 10 of 20; broken areas 20 of 45.	Per photo suggestions only. Dent detection is disabled; a technician confirms counts.
Amazon VisA capsules	Flower found 13 of 20 anomalies and falsely flagged 61 of 121 normal photos.	Earlier manufacturing exercise; its weights do not inspect phones or automotive clips.

The curriculum comparison suite uses handwritten digit images and generated numerical sequences to exercise classification, clustering, dimensionality reduction, forecasting and neural networks. Those results prove the code paths run; they do not measure factory or phone accuracy.

7 Federated learning explained

Federated learning starts two compatible clients with the same model design. Each client trains on its own assigned examples, returns model parameters and its training count, and a coordinator averages compatible parameters with more weight for a client that trained on more examples. The combined model is then evaluated before any approval. A round is one cycle of local training and combination.

ForgeCare has completed real Flower FedAvg training on Renault AutoVI pipe clips and cast pump impellers, with three rounds each. In the automotive run, two clients used 179 photos each. In the casting run, two clients used 455 each. The Federation page shows round start and finish times, participation and final held out worker counts. Both clients ran inside this one laptop. This verifies the algorithm, but does not prove that two independent factories have exchanged weights or that parameter sharing alone protects privacy.

The phone schedule is separate from those completed studies. Each exact phone model needs at least 24 eligible, quality passed, reviewed photos and at least six different acceptable plus six different damaged physical phones. Several views of the same phone still represent one sample. If the same sample ID has contradictory overall labels, it is excluded from training. On the last checked attempt, 6 November 2026 at 17:41 IST, the iPhone had nine reviewed photos but no eligible training set and one conflicting phone ID; the realme had no reviewed photos. Both were skipped, so no phone specific combined model was produced in that attempt.

8 Knowledge assistant and agent

The assistant indexes approved sources rather than searching arbitrary web pages during a worker question. It first filters information by site, subject and source scope, then retrieves relevant passages. Where a supported language model is available, it drafts a conversational answer with citations; otherwise it can present relevant source excerpts. A bounded agent may gather asset facts and approved guidance before drafting an inspection task, but it cannot approve its own task or control machinery.

The prepared phone sources include Apple's iPhone 14 Plus repair and broken glass guidance and realme specifications and support material. Approved general sources also cover selected manufacturing and automotive quality topics. A specification sheet can answer product facts but should not be treated as a repair procedure. The user must open the cited source before relying on a consequential instruction. Groq is optional and its text and visual drafts still require human checking.

9 Architecture and data handling

Component	Current implementation
Operations service	Python FastAPI and SQLite manage users, assets, cameras, captures, labels, work orders, rules, approvals and audit events.
Knowledge service	Separate Python service stores approved documents, retrieves passages and controls cited text generation.
Camera relay	A paired HTTPS path accepts Samsung IP Webcam photos sent by a bridge running on a laptop sharing the phone's Wi Fi.
Dashboard	Python Streamlit provides the worker and supervisor interface. Browser camera permission belongs to the viewing device.
Research and training	Model artifacts, split manifests, source hashes and metrics are saved separately from operational records.
Public access	A fixed Render gateway address forwards to this laptop's live tunnels. The database and images stay on the laptop; it must remain powered and connected.

Operational records and captured images are saved under this laptop's ForgeCare data directory, outside the code package. Public research images and model runs are under the project work and research directories. The portable ZIP does not automatically carry the original research datasets or this laptop's live database. No API token should be placed in a presentation or public repository.

10 Live demonstration sequence

1. **Open the stable address.** Use https://forgecare-ai-live.onrender.com and sign in. The gateway may take around a minute to wake after an idle period. The host laptop and its two tunnels must be running.
2. **Show the two subjects.** In Assets and setup, point out the separate iPhone 14 Plus and realme narzo 60 5G records. Explain that one physical phone per model is enough to demonstrate the workflow but not enough for independent model training.
3. **Show a genuine camera.** Use a viewing device's browser camera with permission, or use the Samsung IP Webcam bridge on a laptop sharing the Samsung's Wi Fi. The TCS emulator may block camera and local network access; the physical laptop must test the Samsung IP stream.
4. **Capture and review.** Take two or more clear views. Show the actual images, quality result, candidate overlays and editable draft. Correct any incorrect AI wording and confirm the physical finding rather than presenting the draft as ground truth.
5. **Follow the case.** Save a clear review, open its inspection order, explain the priority reasons, assign it, approve the instructions, record actual work and mark the order Completed. Show the corresponding Audit entry.
6. **Show supporting evidence.** Ask the Knowledge assistant a source backed question. Then show the completed automotive Flower timeline, the missed and false alert counts, and the phone schedule's honest waiting state.

11 Known limitations and next pilot

- The automotive research models have substantial false alerts and are not approved to release or reject physical parts on a production line.
- No public phone study establishes scratch or dent counting accuracy on the demonstration iPhone or realme. The local dent detector failed its held out test and is disabled.
- The current Flower clients share one laptop. Separate site transport, secure aggregation and privacy claims need their own implementation and evaluation.

- The Raspberry Pi 3B has not been validated as a camera or inference device in this prototype. The working live setup is laptop based.
- The public address is stable, but the active app is hosted by the user's laptop. Power, network, tunnel and free gateway availability remain dependencies.
- Factory inspection duration, cost savings, defect prevalence and employee usability have not been measured in a customer pilot.

A sensible pilot would collect multiple independently identified parts and phones per condition, across operators, lighting and days. It would define defect labels with a domain expert, compare the AI assisted queue with the existing inspection process, measure missed defects and unnecessary rechecks, and test cited guidance against an approved question set. Release decisions should use pre-agreed site specific gates and a human sign off. Until then, ForgeCare is a traceable prototype and research workbench.

12 Questions likely to arise in a review

Why use phones in an automotive project

Phones are available physical subjects for a live capture, review and work order demonstration. Renault AutoVI pipe clips provide the automotive image study. The phone results do not validate vehicle manufacturing inspection.

Is the federated training real

Yes for the public automotive and cast part studies: Flower clients trained model weights over three rounds and saved combined models. Both clients operated on this laptop, so cross factory federation has not been demonstrated.

Why not automatically accept the AI result

A missed defect could let a bad part pass, while a false alarm consumes inspection time. Current study results show both errors. Poor images and generated descriptions add uncertainty. A technician confirms the finding and a supervisor controls the work order.

What is the business value today

The present value is a demonstrable, auditable process for collecting evidence, ordering work and finding approved guidance. Financial savings and improved defect detection are pilot hypotheses, not measured outcomes.

A short explanation you can present

ForgeCare AI addresses a common problem in small manufacturing teams: cameras can produce many observations, but teams have limited time and need evidence before deciding what to inspect. Our prototype starts with a real photo. It checks image quality, allows several views of the same physical object and gives a technician an editable AI draft. Once the technician confirms the visible condition, ForgeCare opens a work order, explains its priority and helps a supervisor allocate the available inspection time. The worker can ask a source linked assistant for approved guidance, then records the actual inspection and completion. Those reviewed outcomes form a traceable basis for future model improvement.

We also trained research models on real automotive pipe clips and manufactured cast parts. The Flower runs completed three rounds of model weight aggregation using two data partitions on one laptop. Their held out results show both caught defects and substantial false alarms, so we do not present them as ready for production. The phone demonstration validates the workflow, while phone specific training waits for enough different reviewed devices. Our next step is a controlled factory pilot that measures missed defects, unnecessary rechecks, worker time and source correctness.

Terms to explain during the demo

Term	Plain meaning
Capture	One saved photo with a camera, subject, time and quality result.
Physical sample ID	The identity of the actual phone or part; several photos can belong to one sample.
Technician review	The human checked description and condition label saved after inspecting the images or object.
Work order	A task that moves through review, assignment, approval, actual inspection and completion.
False alert	A good item sent for an unnecessary extra check.
Federated round	Clients train compatible models separately and combine their parameter updates once.

Where to find the evidence

The application folder on the prepared laptop is C:\Users\crazy\Documents\Codex\2026-11-03\ba\outputs\ForgeCareAI-Release. Its docs\RESEARCH_AND_DATA.md records exact source links, image counts, test splits and training commands. The research folder contains saved model weights, result files and selected examples; the separate work\datasets folder contains downloaded public images. Live captures, work orders, users and audit events live in the private ForgeCare data directory rather than inside the portable code ZIP.

For the automotive proof, show research\autovi_pipe_clip_20261105 for the normal only model and research\autovi_federated_20261105 for the three round Flower run. For the cast part study, use research\casting_impeller_20261105. The Federation screen provides a simpler business view of each completed run and the current phone readiness. A saved model file by itself does not prove accuracy; use its matching split manifest and held out results.

Checks before presenting

- Open the fixed public address and wait for the free gateway to wake. Check that the host laptop and local ForgeCare services are running.
- For a live camera, confirm that the picture changes now. If using Samsung IP Webcam, the phone and physical bridge laptop must share Wi Fi and have a current pairing code.
- Use a clear photo and say which outputs are technician confirmed, model suggestions, research results or fictional practice data.
- Keep a reviewed Apple or realme source question ready, plus the completed automotive study and its false alert count. Do not claim that the phone model has already completed its six hour training gate.

Sources and project records

Primary project brief: Federated Equipment Health BRD version 3.1, supplied by the team on 3 November 2026. Implementation and acceptance notes: ForgeCareAI-Release/README.md, docs/ACCEPTANCE.md, docs/RESEARCH_AND_DATA.md and docs/DEMO_SUBJECTS.md. The research note includes dataset licenses, source links, split counts, training commands and saved artifact paths. Public source families: Renault AutoVI via Zenodo, cast pump impellers and phone photos via Kaggle, phone exterior boxes via Roboflow Universe, and Amazon VisA. The dataset owners' terms should be checked before redistribution or commercial use.